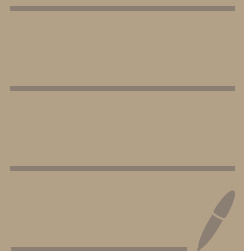


Lec 13. Nonlinear stability of equilibrium

- Ref. [BDM] 9.3 (no proof)
- Ref: [T], Sec 3.7.
(a bit advanced).



I Nonlinear stability:

Consider

$$\textcircled{1} \quad \frac{d}{dt} x(t) = F(x(t)), \quad F(x^*) = 0$$

When x^* is stable / Asymptotic stable??

Done $F(x) = A \cdot x$.

Example: $\frac{d}{dt} x = -x + x^2$, $x^* = 0$ equilibrium.
 $x^* = 1$

stability of $x^* = 0$ ✓ stable
 $x^* = 1$ unstable.

Linearization around x^* .

$$F(x) = \vec{F}(x^*) + \underbrace{\nabla \vec{F}(x^*)}_{A \rightarrow \text{Jacobian matrix}} \cdot (x - x^*) + \underbrace{G(x - x^*)}_{= o(x - x^*)}$$

Let $z = x - x^*$. Then $\textcircled{1} \Rightarrow$.

$$\textcircled{2} \quad \frac{d}{dt} z(t) = A \cdot z(t) + G(z), \quad z_0 = x_0 - x^*$$

From Taylor expansion, we know.

$$\textcircled{2.6} \quad |G(z)| \leq M \cdot |z|^2, \quad \text{here } |\vec{v}| = \text{norm } |v| = 1. \quad \downarrow \text{vector norm.}$$

$\forall |z| \leq 1.$

Heuristic $\textcircled{2} \approx$

$$\frac{d}{dt} z(t) = A \cdot z(t) + \text{small.}$$

Thm (Nonlinear stability).

- Suppose F is C^1
- Denote $A_{ij} = \partial_i F_j(x^*)$.
- Let λ_i be the eigenvalues of A

Ass If $\operatorname{Re}(\lambda_i) \leq -\lambda < 0$, $\forall i$, then x^* is asymptotically stable. i.e.

$$|x(t) - x^*| < \delta \quad \text{for } \delta \text{ small enough.}$$

$$\lim_{t \rightarrow \infty} |x(t) - x^*| = 0.$$

Remark: Assumption can weaken to FGC!

pf: Step 1: Existence and uniqueness $\Rightarrow$ solution exists $t \in [t_0, t_1]$.

Step 2: Linearization

(Ass) $\Rightarrow \frac{d}{dt} z(t) = A \cdot z + \text{small}$

$$\approx A \cdot z$$

is stable.

Step 3: Variation of constant. (pretend $G(z)$ is known)

$$(3) \quad z(t) = e^{At} z_0 + \int_0^t e^{A(t-s)} \underbrace{G(z(s))}_{\text{want small}} ds.$$

recall $|e^{At} v| \leq c e^{-\lambda t} t^n \cdot |v|$ last class

$$(4) \quad < a \cdot e^{-\frac{2}{3}\lambda t} |v|$$

Want to say that $z(t)$ behavior like $\boxed{e^{-\frac{2}{3}\lambda t}}$

so consider

$$d(t) = e^{\frac{2}{3}\lambda t} |z(t)|, \text{ show } d(t) \text{ small}$$

Goal: For any ε small enough, there exists $\delta > 0$.

$$\text{s.t.} \quad \text{If } |z_0| < \delta \Rightarrow |d(t)| < \varepsilon, \quad \forall t \geq 0$$

Step 4 :

$$\begin{aligned} d(t) &= e^{\frac{2}{3}\lambda t} |z(t)| \\ &\leq e^{\frac{2}{3}\lambda t} |e^{At} z_0| + e^{\frac{2}{3}\lambda t} \int_0^t |e^{A(t-s)} G(z(s))| ds \\ &= I + II \end{aligned}$$

$$\textcircled{4} \Rightarrow |I| \leq e^{\frac{2}{3}\lambda t} \cdot a \cdot e^{-\frac{2}{3}\lambda t} |z_0| < a \cdot \delta$$

$|G(z(s))| \leq M \|z(s)\|^2$ is true
since we will show $z(s)$ is small.

②. 6 + ④

$$\begin{aligned} \Rightarrow II &\leq e^{\frac{2}{3}\lambda t} \int_0^t e^{-\frac{2}{3}\lambda(t-s)} \cdot a |G(z(s))| ds \\ &\leq e^{\frac{2}{3}\lambda t} \int_0^t e^{-\frac{2}{3}\lambda(t-s)} a M \|z(s)\|^2 ds \\ &= aM \int_0^t e^{\frac{2}{3}\lambda s} \cdot \|z(s)\|^2 ds \\ &= aM \cdot \int_0^t e^{-\frac{2}{3}\lambda s} (e^{\frac{2}{3}\lambda s} \|z(s)\|^2) ds \\ &= aM \int_0^t e^{-\frac{2}{3}\lambda s} (d(s))^2 ds. \end{aligned}$$

So

$$\textcircled{5} d(t) \leq I + II \leq a\delta + aM \int_0^t (d(s))^2 e^{-\frac{2}{3}\lambda s} ds$$

When can we say $d(s) < \varepsilon$, $SG[0, t) \Rightarrow d(t) < \varepsilon$?

• Induction on time (Bootstrap).

• Base case: $t=0$ $d(0) = |z_0| \leq \underline{\delta} < \varepsilon$.

• Induction hypothesis: $d(s) < \varepsilon, s \in [0, t)$. constraint.

• Want $d(t) < \varepsilon$. (By continuity $\Rightarrow d(s) < \varepsilon, s \in [0, t + \alpha]$)

$$\begin{aligned} d(t) &\leq a \delta + aM \cdot \int_0^t \varepsilon^2 e^{-\frac{2}{3}\lambda s} ds \\ &\leq a \delta + aM \cdot \varepsilon^2 \cdot \frac{1}{\frac{2}{3}\lambda} \end{aligned}$$

If

(6)

$$\boxed{a \delta + \frac{3aM \varepsilon^2}{2\lambda} < \frac{\varepsilon}{2}}$$

$$\Rightarrow d(t) < \frac{\varepsilon}{2} < \varepsilon.$$

Step 5:

(6) is possible. If $\frac{3aM \varepsilon^2}{2\lambda} < \frac{\varepsilon}{2}$

$$\Leftrightarrow \varepsilon < \frac{\lambda}{6aM}$$

then when

$$|z_0| < \delta < \frac{\varepsilon}{4a}.$$

we get (6)

² Induction on time work!! $\Rightarrow d(t) < \varepsilon$

$$\Rightarrow |z(t)| < \varepsilon \cdot e^{-\frac{2}{3}\lambda t} \rightarrow 0$$

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. IV: Application.

Ex1. $\frac{d}{dx} x = ax^2 + bx + c.$

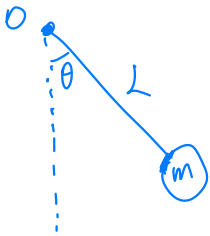
Ex2. 2×2 system. (books).

Motion of pendulum.

$$\frac{dx}{dt} = y, \quad \frac{dy}{dt} = -\omega^2 \sin x - \gamma \cdot y.$$

$y=0$, $x=k\pi$

$\Rightarrow \frac{d^2\theta}{dt^2} + \gamma \frac{d\theta}{dt} + \omega^2 \sin\theta = 0$



unstable.